Generate

Digital Signal Intelligence
Revolutionary Technical Pricing Model

Using enterprise digital footprint analysis to transform underwriting economics and risk selection

THE CORE THESIS

A company's digital presence—how they maintain their website, who they're connected to online, and what they disclose—serves as a powerful proxy for operational maturity, governance quality, and risk management capability. By systematically scoring these "digital signals," we can automate 75-85% of underwriting decisions while simultaneously improving risk selection, enabling zero-touch pricing for most submissions.

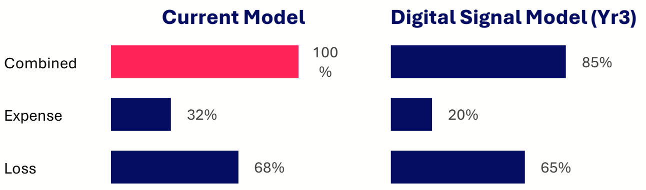
This isn't about adding automation to existing models, it's about replacing manual underwriting with an entirely new data pillar that works globally, updates continuously, and costs 90% less than traditional methods.

5-YEAR FINANCIAL IMPACT

$275 – 300m
Cumulative profit improvement

26 – 34pts
Combined ratio improvement

1800%+
Return on investment



HOW DIGITAL SIGNALS WORK

We extract and quantify four signal categories from public web data:

- **Infrastructure signals**
  SSL certificates, security posture, technology stack (predicts cyber exposure)
- **Content signals**
  Governance transparency, disclosure depth, policy visibility (predicts D&O risk)
- **Network signals**
  Link quality, vendor ecosystem, supply chain connections (predicts credit/political risk)
- **Behavioural signals**
  Maintenance discipline, incident response speed, operational consistency (predicts all lines)

D&O
12 – 18pt LR improvement

Cyber
15 – 25pt LR improvement

E&O
10 – 15pt LR improvement

Credit
Early warning system

Marine
Risk segmentation

OPERATIONAL TRANSFORMATION: FROM 2 WEEKS TO 4 HOURS

	Current state	Digital signal model
Timeline	2 -14 days	45 mins – 4 hours
Touch-points	8 – 15 manual steps	0 –2 (exceptions only)
Cost per quote	$650	$70
Hit-ratio	35%	45%
Note	Manual data gathering, surveys, underwriter review, multiple approval layers	Automated lookup, instant risk scoring, rule-based decision making, 75 – 85% straight-through

$25.7M Annual Expense Savings
On 50,000 submissions (typical mid-carrier market volume)

PHASED IMPLEMENTATION: RE_RISKED APPROACH

Phase 1
Proof of concept
D&O Validation
6 months
$0.8 – 1.2M

Phase 2
Multi-coverage
Cyber, EO&O, D&O, Credit
6 months
$1.5 – 2.2M

Phase 3
Full automation
Global deployment
12 months
$2.5 – 3.5M

Phase 1 success criteria

- Digital score predicts loss frequency
- Top quintile shows 30%+ lower loss ratio
- 90%+ company coverage achieved
- Statistical significance validated

Data Sources (70%+ coverage)

- Common Crawl (free web archive)
- Certificate Transparency logs
- Ahrefs/SEMrush (link analysis)
- Proprietary scoring algorithms

WHY THIS MATTERS NOW

- **First-Mover Window**
  18-36 months before competitors replicate; regulatory approvals create barriers
- **Market Expectations**
  Instant quotes are table stakes in 2025; 2-week turnarounds lose business
- **Talent Magnet**
  Talent won't join "insurance as usual" companies; need cutting-edge projects
- **Margin Compression**
  Without operational gains, target ROE becomes unachievable in competitive markets

DEFENSIBLE COMPETITIVE ADVANTAGES

- **Data Moat**
  Proprietary historical validation and signal calibration (3+ years to replicate)
- **Network Effects**
  More scored companies = more accurate relative positioning
- **Regulatory Lead**
  Rate filings and actuarial documentation create 2–3-year barriers
- **Learning Loop**
  Claims feedback improves model faster than competitors starting from scratch

BOTTOM LINE FOR THE BOARD

Approve phase 1 – proof of concept
$800K - $1.2M
6-month validation of D&O coverage – definitive evidence of predictive power, decision point before commitment

Success outcome
Fast-track phase 2
Immediate competitive advantage

Neutral outcome
Refine and retry
Limited downside, learning value

Failure outcome
Stop investment
$1M loss, clear market signal

Success Probability
70-80% that digital signals show statistically significant predictive power for 2-3+ coverage lines based on academic research on organisational digital maturity and preliminary data exploration

Risk-Adjusted Expected Value
$175-280M (70% probability × $250-400M potential 5-year value)